



EMBENTION

Fabrication notes
hw_DAA/BASIC-101010

September 24, 2024

1. PCB construction

- Number of layers: 16 layers.
- Number of flex layers: 6 (3 core structures joint with polyamide bondply).
- A strain relief shall be added over the flex and rigid intersection.
- Surface finish: Immersion silver.
- PCB Material requirement:
 - High TG material, greater than 180°C.
 - RoHS compliant.
 - REACH compliant.
- Dielectric material: must comply with IPC-4101-126 standar.
- Soldermask: on both sides of the board, color green.
- All vias in pads shall be plugged with solder mask type V/VI.
- Layer stack shall comply with the IPC-6012C.
- Layout clearance shall comply with the IPC-2221B.
- PCB shall comply with the IPC 6011 Class 3/A.

2. Gerber description and stackup

The layer structure is defined below:

1. Layer Top	→	*.cmp
2. Layer 2	→	*.l2
3. Layer 3	→	*.l3
4. Layer 4	→	*.l4
5. Layer 5	→	*.l5
6. Layer 6	→	*.l6
7. Layer 7	→	*.l7
8. Layer 8	→	*.l8
9. Layer 9	→	*.l9
10. Layer 10	→	*.l10
11. Layer 11	→	*.l11
12. Layer 12	→	*.l12
13. Layer 13	→	*.l13
14. Layer 14	→	*.l14
15. Layer 15	→	*.l15
16. Bottom	→	*.sol
17. Solder mask top	→	*.stc
18. Solder mask bottom	→	*.sts
19. Dimensions	→	*.dim
20. Panel	→	*.gml
21. Milling	→	*.gko
22. Flex	→	*.flx
23. Cream top	→	*.tcr
24. Cream bottom	→	*.bcr
25. Silkscreen top	→	*.tsilk
26. Silkscreen bottom	→	*.bsilk
27. Drills	→	*.drd

The PCB stackup is described in the next image:

1		Nr	Copper	Isolation
2		1	0.042mm	0.13mm
3		2	0.017mm	0.1mm
4		3	0.017mm	0.13mm
5		4	0.017mm	0.1mm
6		5	0.017mm	0.13mm
7		6	0.017mm	0.05mm
8		7	0.017mm	0.1mm
9		8	0.017mm	0.05mm
10		9	0.017mm	0.1mm
11		10	0.017mm	0.05mm
12		11	0.017mm	0.13mm
13		12	0.017mm	0.1mm
14		13	0.017mm	0.13mm
15		14	0.017mm	0.1mm
16		15	0.017mm	0.13mm
		16	0.042mm	
		Total:		1.852mm
Setup	(1+2*3+4*5+6*7+8*9+10*11+12*13+14*15+16)			

3. Drills

Number of drills: 3241

Drills used:

Code	Size	Used
T01	0.0059inch	2874
T02	0.0079inch	61
T03	0.0098inch	17
T04	0.0197inch	154
T05	0.0291inch	32
T06	0.0315inch	68
T07	0.0394inch	4
T08	0.0591inch	1
T09	0.0669inch	4
T10	0.1181inch	4
T11	0.1299inch	4
T12	0.1575inch	12
T13	0.2205inch	6

5. Silkscreen

The PCB shall have silkscreen in the layers top and bottom.

The gerber data for silkscreen are the *.tsilk and *.bsilk.

Do not include manufacturing logos in the PCB.

The silkscreen color shall be white.

PCB manufacturers shall add a traceability number in the products. This number shall be the PCB batch and a week-year number.

6. Additional notes

The PCB manufacturing shall comply with the IPC-1602 standard to ensure a correct handling, protection and storage.

Two additional documents are added for the manufacturer:

- hw_DAA_Rigiflex_drawing.pdf: Describes placement and dimensions of the rigiflex segments.
- hw_DAA_Drill_and_Countersunk_drawing.pdf: Describes the placement and size of the countersunk drills.